

Phage Identification and Phage-host Interaction Prediction Using k-mers and Machine Learning

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Introduction

Metagenomics

The study of genetic material recovered directly from environmental samples

- The Global Ocean Survey (Venter et al. Science 2004)
- The Human Microbial Project (HMP)
- The MetaHIT (human intestinal track) project
- The Earth Microbiome Project
- The TARA Ocean Project
- US National Microbiome Initiative
- US NIH Human Virome Project

The world of a drop of seawater

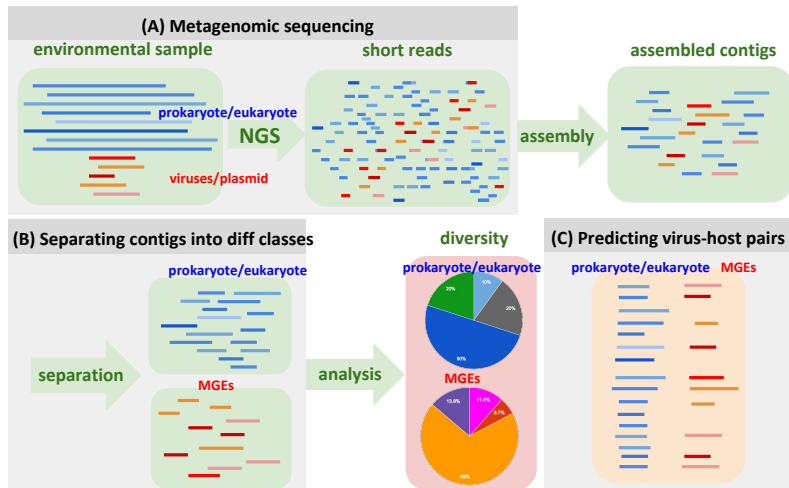
- 10 M viruses, 1 M bacteria/archaea, 1 K small protists (protozoans and algae), plasmids, etc
- Viruses: the most abundant biological entities, but less is known due to the difficulty of virus isolation
- Plasmids can impact the survival of their hosts
- Most current studies concentrate on prokaryotes with little attention to MGEs



picture from www.dreamstime.com

http://www.marbef.org/wiki/Microbial_research

Studying MGEs using metagenomic data



Different types of metagenomic experiments

- **Cellular metagenomes:** Sequencing all microbial organisms without filtration. All microbes including viruses, fungi, bacteria, archaea, etc are present
- **Size-fractionated metagenomes:** filtration of certain types of microbes such as eukaryotes, prokaryotes, viruses, etc

In both scenarios

- there potentially are both MGEs and other microbes including bacteria, archaea, etc
- the relationship between MGEs and their hosts are lost

Objectives

- Identification of phage contigs
- Predict phage-host associations (interactions)

VirFinder and DeepVirFinder: Identification of phage contigs from metagenomes

Publications related to virus identification and applications

Ren et al. *Microbiome* (2017) 5:69
DOI 10.1186/s40168-017-0283-5

Microbiome

METHODOLOGY

Open Access



VirFinder: a novel *k*-mer based tool for identifying viral sequences from assembled metagenomic data

Jie Ren^{1*}, Nathan A. Ahlgren^{2,4*}, Yang Young Lu¹, Jed A. Fuhrman² and Fengzhu Sun^{1,2*}

Bioinformatics, 37(8), 2021, 759–766

doi: 10.1093/bioinformatics/btae812

Advance Access Publication Date: 29 October 2021

Original Paper

OXFORD

Sequence analysis

KIMI: Knockoff Inference for Motif Identification from molecular sequences with controlled false discovery rate

Xin Bai¹, Jie Ren¹, Yingying Fan^{2,*} and Fengzhu Sun^{1,*}

Quantitative Biology 2020, 8(1): 64–77
<https://doi.org/10.1007/s40484-019-0187-4>

RESEARCH ARTICLE

Identifying viruses from metagenomic data using deep learning

Jie Ren^{1,1*}, Kai Song^{2,1}, Chao Deng¹, Nathan A. Ahlgren³, Jed A. Fuhrman⁴, Yi Li⁵, Xiaohui Xie⁵, Ryan Poplin⁶, Fengzhu Sun^{1*}

NAR Genomics and Bioinformatics, 2024, 6, 1qae044

<https://doi.org/10.1093/nargab/1qae044>

Advance access publication date: 6 May 2024

Standard Article

OXFORD

DeepMicroClass sorts metagenomic contigs into prokaryotes, eukaryotes and viruses

Shengwei Hou^{1,2,*}, Tianqi Tang^{2,1}, Siliangyu Cheng^{2,1}, Yuanhao Liu¹, Tian Xia¹, Ting Chen⁴, Jed A. Fuhrman² and Fengzhu Sun^{1,*}

Previous methods and challenges

- **Alignment to known viral reference genomes** (Metavir): only about 10-15% viruses have similarity to the known viruses.¹
- **Viral gene homology based tool** (VirSorter): only sequences contain ≥ 3 genes can be evaluated
 - not suitable for sequences from non-coding regions
 - not suitable for mostly short sequences (80% <3000 bp)

¹Norman et al.(2015). Cell; Hurwitz et al. (2013). PloS One.

A new method should be able to ...

- Identify viral contigs both containing genes or not
- Achieve decent prediction accuracies for short contigs
- Discover new viral contigs no matching to known viruses

VirFinder: k -mer based, machine learning method

IDEA: Viruses and bacterial hosts have different usage of word patterns

- Characterizing sequences using k -mer frequencies regardless of coding and non-coding regions
 - For a sequence S , N_w is the word count. The feature vector is defined as,

$$V_w = \frac{N_w}{\sum_w N_w}, \mathbf{w} \in \mathcal{A}^k.$$

- Binary classification: $Y = 1$ if viruses, $Y = 0$ if hosts
- Logistic regression with lasso regularization

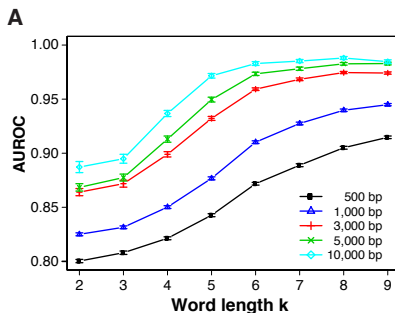
The Dataset

- 1,562 virus & 31,986 prokaryotic RefSeqs (May 2015, NCBI)
- **Fragment** RefSeq into **non-overlapping contigs**
- Same number of virus and host contigs
- **Train** the model using contigs from genomes **before 2014/1/1**
- **Test** the model using contigs from genomes **after 2014/1/1**.

contig length L	Before 2014	After 2014	Total
500 bp	154,640	50,350	204,990
1,000 bp	77,014	25,087	102,101
3,000 bp	25,263	8,246	33,509
5,000 bp	14,881	4,878	19,759
10,000 bp	7,120	2,357	9,477

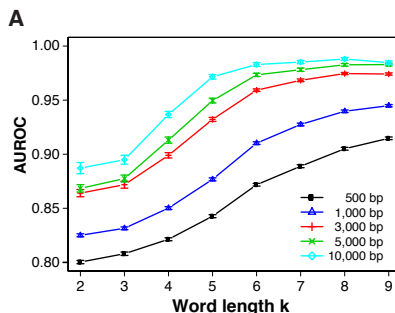
Table: The number of sequences for training and testing

The effects of k -mer length and contig length



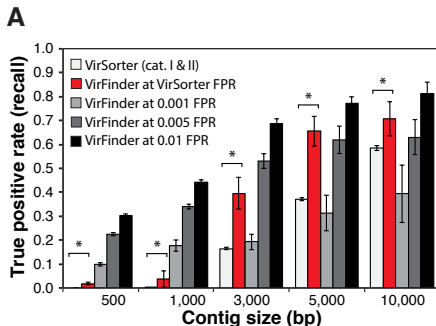
- AUC increases as k -mer length increases; stabilized for $k \geq 8$
 - k -mer length 8 is chosen
- Longer contig, better performance

The effects of k -mer length and contig length



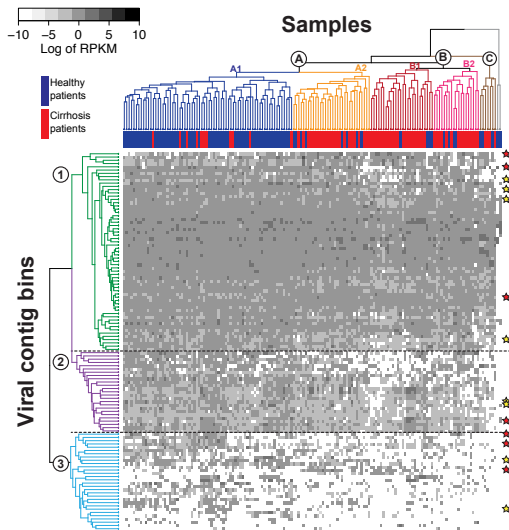
- AUC increases as k -mer length increases; stabilized for $k \geq 8$
 - k -mer length 8 is chosen
- Longer contig, better performance
- VirFinder has 3 models:
 - the 500 bp trained model: contigs $< 1,000$ bp
 - the 1,000 bp trained model: contigs 1,000-3,000 bp
 - the 3,000 bp trained model: contigs $\geq 3,000$ bp

VirFinder identifies more viral contigs than VirSorter



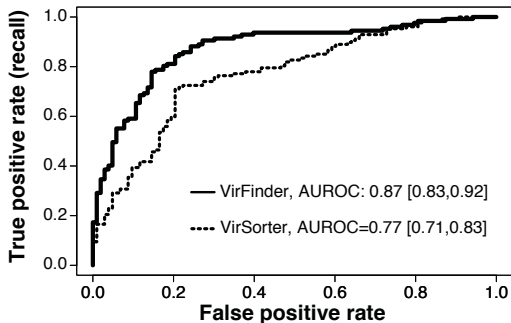
- Compare the fraction of true viral contigs identified by VirFinder (VF) and VirSorter (VS) at the same false positive rate (FPR)
 - For 500 bp, VF identifies 2%, VS 0%, FPR 0
 - For 1,000 bp, VF identifies 4%, VS 0%, FPR 10^{-5}
(VF recalls 18% at FPR 0.001)
 - For 3,000 bp, VF identifies 40%, VS 16% (2.5 fold higher), FPR 0.003

Identifying viruses in human gut metagenomes from a liver cirrhosis study ²



- 2-way hierarchical clustering of viral contig bins and human gut metagenomic samples
- Rows: 116 viral contig bins comprised of 2,657 viral contigs (q-value 15%)
- Columns: 80 healthy (blue) and 76 cirrhosis (red) patients
- Averaged Reads Per Kilobase per Millions mapped reads (RPKM) of contigs in each bin
- Bin cluster (1) in nearly all (95%) samples; Bin cluster (2) and (3) have significantly low prevalence in cirrhosis patient
- Stars: 7 bins that are positively (red); 8 are negatively (yellow) associated with cirrhosis

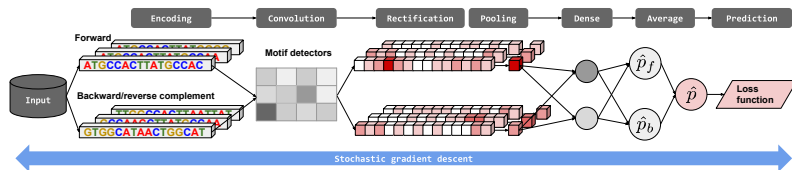
Predicting cirrhosis status using viral contigs



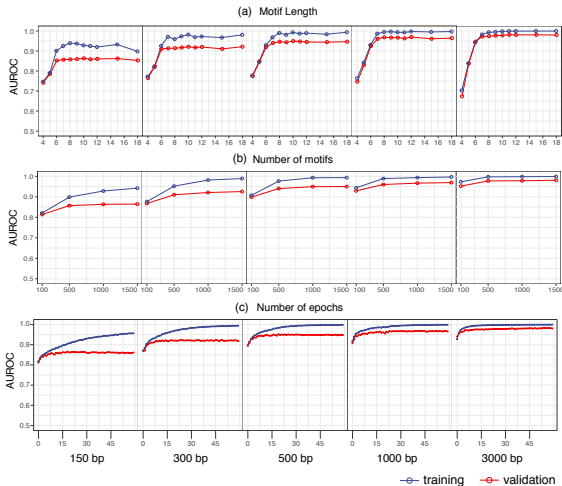
- Learn a prediction model using 38 liver cirrhosis patients and 40 controls with virus contigs using logistic regression with LASSO penalty
- Test the model with an independent set of 230 samples from the same study

DeepVirFinder: Finding viruses using deep-learning (Ren et al. QB 2020, Mo et al. CP 2026)

- Enlarging the virus training database including those from metavirome studies
- Deep learning with nucleotide motifs
- Build models for different contig lengths

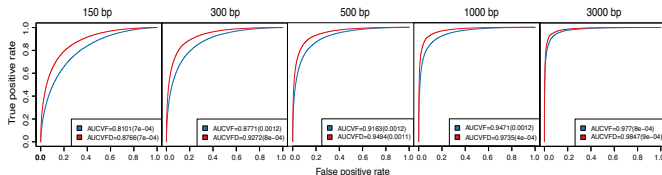


Impacts of motif length, number, and epoch number



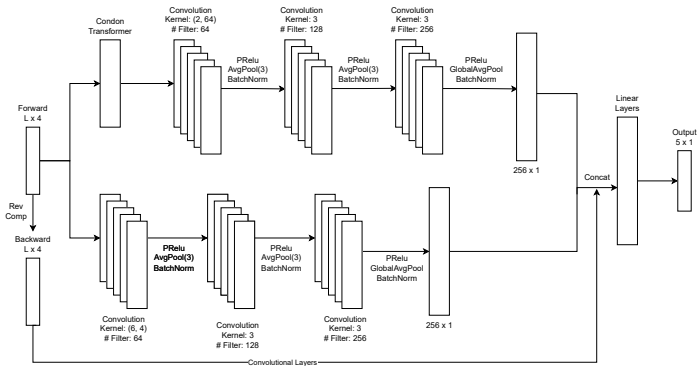
- AUROC increases with motif lengths, stabilize around 8-10
- Increases with number of motifs, stabilizes around 1000

DeepVirFinder outperforms VirFinder



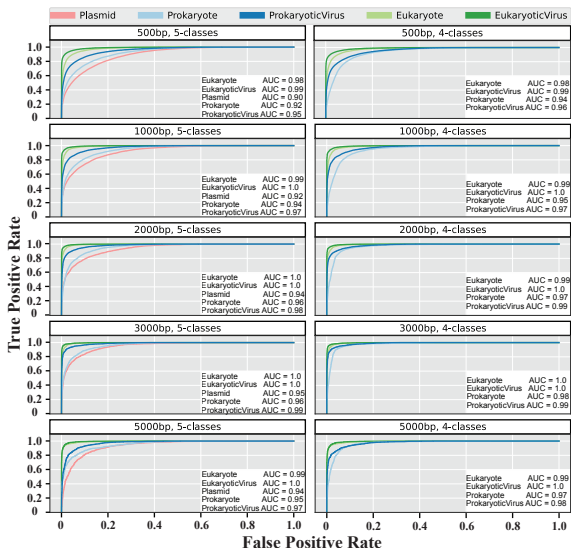
- Updated tutorial available from Current Protocols (2026)
- Updated code at https://github.com/secdio/DeepVirFinder_protocol

DeepMicrobeClass: Classify contigs into multiple classes using deep learning (Hou et al. NARGB 2024)



The CNN network structure for classifying contigs to eukaryotes, prokaryotes, viruses infecting eukaryotes, viruses infecting prokaryotes and plasmids

DeepMicrobeClass multiclass prediction accuracy



VirHostMatcher: predicting virus-host interactions using k-mers (Ahlgren, Ren et al. NAR 2017)

What hosts viruses infect?

- Traditional experiments: isolate viruses using cultured host strains
 - ✓ direct and accurate
 - ✗ low throughput, only for lytic viruses, only for culturable hosts (<5%)
- Metagenomic sequencing: capture all DNA molecules in microbial communities
 - ✓ high throughput, culture-independent
 - ✗ no direct links between viruses and hosts

Computational approaches to infer hosts of viruses

- Sequence alignment: viruses and hosts share genes due to horizontal gene transfer; host genomes may contain genomes of lysogenic viruses.
 - ✓ highly accuracy
 - ✗ fails when viruses and host have no shared genes, e.g., crAssPhage

Alignment-free methods to infer hosts of viruses

- **BASIC IDEA:** Viruses and their host genomes share more similar patterns in k -mers than between viruses and non-infecting hosts
 - viruses utilize the translational functions in the hosts (Carbone, 2008)
 - viruses and hosts are under the same evolutionary pressure (Robins et al., 2005)
- The host of a given virus can be predicted as the one with the smallest distance in k -mers.

What distance/dissimilarity measure to use (Ahlgren, Ren et al. NAR 2017)?

We collect 11 different measures based on two different principles:

- Distance using observed k -mer frequencies N_w :
e.g., Euclidean or Manhattan (commonly used with tetramers)
- Distance using background-adjusted k -mer frequencies $\tilde{N}_w$:
e.g., d_2^* , d_2^S , *CVTree*

We answer the following questions:

- Which measure has the highest prediction accuracy?
- Should we use tetramers (4-mers) or other word length?

Alignment-free dissimilarity measures d_2^* and d_2^S (Reinert et al. 2009, Wan et al. 2010, Ren et al. 2016)

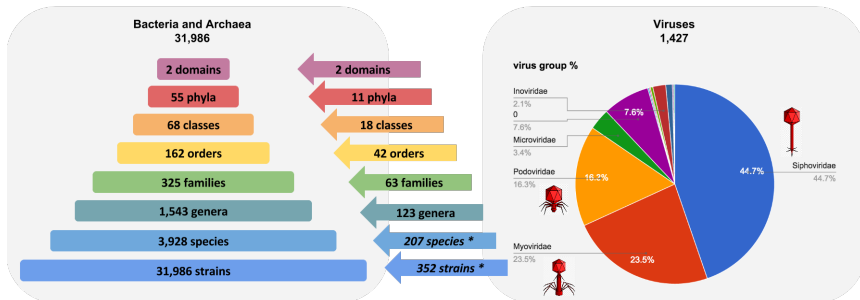
Given two sequence data sets X and Y

- Count the number of occurrences of every k -mer $\mathbf{w}$, $X_{\mathbf{w}}$ and $Y_{\mathbf{w}}$
- Calculate their expectations $E(X_{\mathbf{w}})$ and $E(Y_{\mathbf{w}})$ under certain Markov models defined by the data
- Subtract the expectations from the actual counts and normalize with d_2^* and d_2^S using different normalization procedures
- Calculate the correlation between the normalized counts
- Change to dissimilarity measures

Excellent performance in applications of d_2^* and d_2^S

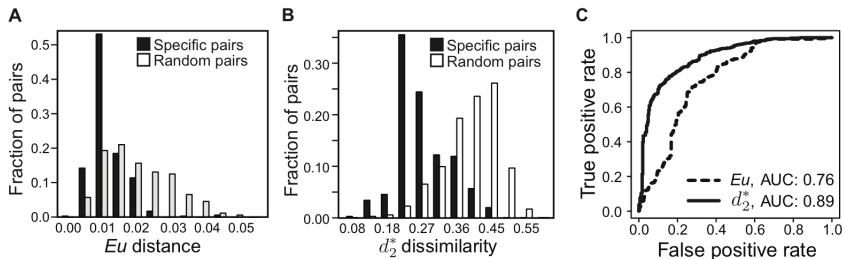
- Estimating evolutionary distance (Ren et al. 2016)
- Evolution of cis-regulatory modules (Song et al. 2013)
- Comparison of metagenomes (Jiang et al. 2012)
- Locating horizontal gene transfer (Tang et al. 2018)
- Finding geographic locations of oak trees (Tang et al. 2018)
- User friendly software CAFE (Lu et al. 2017)
- Review on alignment-free sequence comparison (Ren et al. 2018)

Dataset



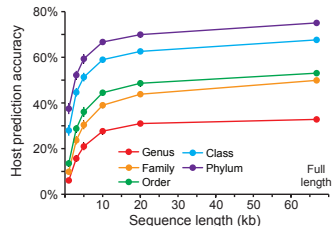
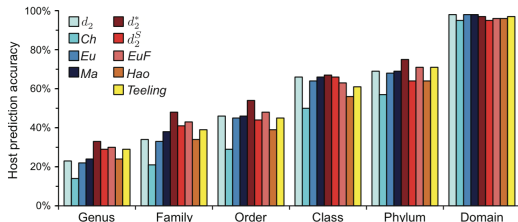
- hosts are biased to certain types
 - bacteria 31,530 - archaea 456
- viruses are biased to certain types of hosts
- 352 one-one virus-host genome pairs

d_2^* outperforms others for predicting virus-host interactions



- 352 interacting pairs - 352 random pairs; 6mers, 2nd order MC
- Measures based on normalized frequencies perform better
- d_2^* performs the best

Host prediction accuracy using different measures



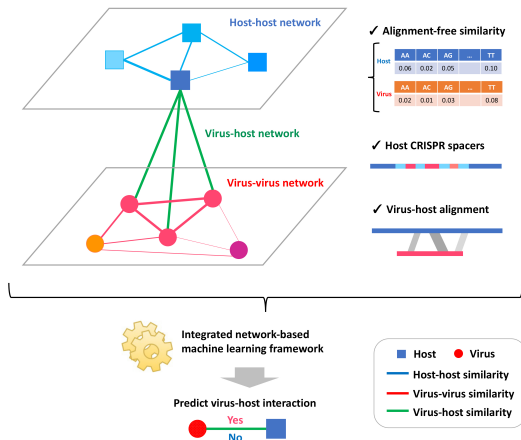
- d_2^* has the highest prediction accuracy; 6mer, 2nd order MC,
 - 33% at genus, 75% at phylum
- Viral genomes are subsampled to study the effect of sequence length on prediction accuracy
 - accuracy is roughly stable down to 5 kb

Other information for virus-host interactions

- Sharing of CRISPR spacers and other regions between the virus and the hosts measured through alignment
- Alignment-free similarity measure $s_2^* = 1 - d_2^*$
- Network Information
 - Similar viruses tend to infect similar hosts
 - Similar bacteria tend to be infected by the same viruses

Integration Framework

- Define the virus-virus and host-host networks by s_2^*
- Incorporate biological signals and alignment-free measures



VirHostMatcher-Net: A network model for virus-host interactions (Wang et al. NARGAB (2020))

$$\text{logit}(\Pr(I(v, b) = 1)) = \alpha + \beta s_2^*(v, b) + \gamma_+ SV_+(v, b) + \gamma_- SV_-(v, b) + \eta S_{CRISPR}(v, G_b) + \delta S_{blastn}(v, b)$$

where

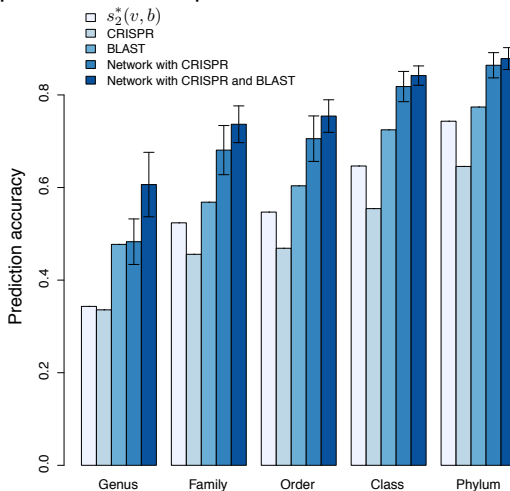
$$SV_+(v, b) = \frac{1}{||H_b||} \sum_{v' \in H_b} s_2^*(v, v'), \quad SV_-(v, b) = \frac{1}{||H_b^c||} \sum_{v' \in H_b^c} s_2^*(v, v'),$$

$$H_b = \{v' | I(v', b) = 1\}, \quad H_b^c = \{v' | I(v', b) = 0\}$$

$S_{CRISPR}(v, G_b)$ and $S_{blastn}(v, b)$ measure the similarity between virus-host based on CRISPR and alignment, respectively

Host prediction accuracies under different taxonomic ranks

Significantly improves the host prediction accuracies



Conclusions

- VirFinder and DeepVirFinder can accurately predict viral contigs, in particular, for relatively short ones
- DeepMicrobeClass can accurately classify contigs into multiple categories
- VirHostMatcher using d_2^* is powerful for identifying virus-host interactions
- VirHostMatcher-Net integrates alignment-free, alignment, network, and CRISPR information for accurately predicting virus-host interactions
- Software available on Github

Key References

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- Dr. Shengwei Hou, Yuqian Mo, Southern University of Science and Technology, China

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